# Integrating Text-Based Problem Descriptions with Data-driven Partial Differential Equation Solver

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## Abstract

Data-driven Partial Differential Equation (PDE) surrogate models can be fast and effective, but they often struggle to generalize beyond the training distribution. This work studies whether natural language descriptions of physical systems can provide useful high-level context for improving such generalization. We extend a U-shaped Neural Network (UNet)-based surrogate model by conditioning it on text embeddings derived from structured PDE descriptions, and we compare this approach against an unconditioned UNet and transformer-based Axial Vision Transformer (AViT) baselines.

Experiments train on turbulent radiative layer and active matter datasets and evaluate out-of-distribution performance on Helmholtz staircase. Across rollout-based metrics, the baseline UNet remains the strongest model, while the conditioned UNet does not consistently improve accuracy and the transformer baseline performs worse. These results suggest that simple text conditioning is not enough to improve out-of-distribution (OOD) generalization in this setting, and that the success of multimodal PDE modeling depends strongly on how semantic information is integrated into the architecture.

## 1 Introduction

Partial differential equations (PDEs) underpin many physical, biological, and engineering models, but solving them accurately at scale remains computationally expensive. Classical numerical methods are reliable, yet they often require careful problem-specific design and become increasingly costly for high-resolution or many-query settings. This has motivated the use of data-driven surrogate models that learn to predict PDE dynamics directly from simulation data.

Recent deep learning models have shown that surrogate PDE solvers can be both accurate and efficient, especially on structured grid data. At the same time, their performance often depends strongly on the training distribution. Most models operate only on numerical states and are asked to infer all relevant physical structure from those states alone, which makes out-of-distribution (OOD) generalization difficult when the governing regime, boundary conditions, or equation family changes.

Natural language descriptions offer a different source of information. PDE systems are often accompanied by structured textual knowledge about equations, boundary conditions, coefficients, and qualitative behavior. Modern large language models (LLMs) can embed this information into dense vectors, making it possible to condition numerical surrogate models on compact semantic context instead of relying only on raw state trajectories.

This work studies whether such text-based conditioning improves PDE surrogate modeling. We extend a UNet-based architecture with text embeddings derived from structured PDE descriptions and compare it against an unconditioned UNet and transformer-based Axial Vision Transformer (AViT) baselines. The models are trained on turbulent radiative layer and active matter data and evaluated on the OOD Helmholtz staircase dataset using rollout-based error metrics.

Our main finding is negative but informative: simple text conditioning does not improve performance in this setting. The baseline UNet remains strongest, the conditioned UNet does not consistently outperform it, and the reported AViT runs lag behind both. These results suggest that multimodal PDE modeling is less about adding another modality and more about aligning that modality with the inductive bias of the solver.

The main contributions of this paper are:

1. We implement a text-conditioned UNet surrogate model that injects PDE descriptions, encoded with a frozen sentence embedding model, into a grid-based forecasting pipeline.
2. We evaluate this conditioned model against an unconditioned UNet baseline and representative AViT transformer runs in a multi-dataset setting with explicit OOD testing on Helmholtz staircase.
3. We show that lightweight text conditioning does not improve OOD rollout performance in this setup, and we use that negative result to argue that solver inductive bias and fusion design matter more than simply adding semantic metadata.

## 2 Problem Setting and Motivation

This section introduces the problem setting and the architectural motivation for text conditioning. The next section focuses specifically on prior literature and how this project is positioned relative to it.

### 2.1 Partial Differential Equations in Scientific Modeling

Partial differential equations (PDEs) describe how physical quantities such as temperature, velocity, pressure, and concentration evolve over space and time. Classical examples include diffusion, wave propagation, and fluid flow. In this work, one benchmark system is the two-dimensional turbulent radiative layer, whose governing equations are shown in Equation 1.

$$
\begin{array}{rcl}
\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) & = & 0, \\
\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + P\mathbf{I}) & = & \mathbf{0}, \\
\frac{\partial E}{\partial t} + \nabla \cdot ((E + P)\mathbf{v}) & = & -\frac{E}{t_{\text{cool}}}, \\
E = \frac{P}{\gamma - 1}, \qquad \gamma = \frac{5}{3}. & &
\end{array}
\tag{1}
$$

Here $\rho$ denotes density, $\mathbf{v}$ velocity, $P$ pressure, $E$ internal energy density, $\mathbf{I}$ the identity tensor, and $\otimes$ the outer product.

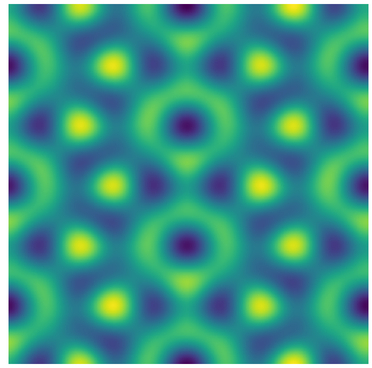
Analytical solutions exist only for a limited class of problems, so most realistic PDEs are solved numerically using methods such as finite differences, finite elements, or spectral discretizations. These methods are accurate and physically grounded, but they are also computationally expensive and often require substantial problem-specific design. Figure 1 shows example fields from the active matter and turbulent radiative layer datasets used in this study.

### 2.2 Data-Driven Surrogates and Generalization

The rise of machine learning has introduced surrogate models that learn mappings from input states to future system behavior directly from simulation data. Once trained, these models can produce predictions much faster than repeatedly running a full numerical solver. Convolutional architectures such as UNets are especially useful for grid-based PDE data because their local connectivity and weight sharing align with spatially structured fields.

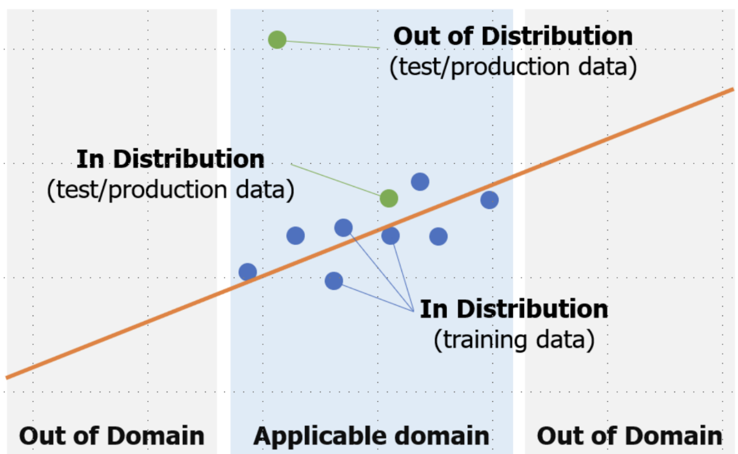
However, data-driven PDE solvers still depend on representative training data. Generating this data often requires computationally expensive simulations, and learned models may overfit to the regimes included in those simulations. This makes generalization a central challenge, especially when a model is evaluated under new parameters, boundary conditions, or governing equations.

Figure 1: Example fields from the datasets used in this study: (a) active matter and (b) turbulent radiative layer.



Generalization is a key requirement for practical PDE solvers. In many applications, a model must operate under conditions that differ from those seen during training, including new initial conditions, parameters, boundary conditions, or even governing equations. Such scenarios are commonly referred to as OOD settings, as illustrated in Figure 2.

Figure 2: Illustration of in-distribution, out-of-domain, and out-of-distribution settings.




Data-driven models often perform well in-distribution but degrade sharply in OOD settings because they learn correlations from the training data rather than broadly transferable physical structure. A model trained on one coefficient range or one type of boundary condition may therefore fail when those assumptions change. This motivates the search for additional signals that can guide models toward more robust behavior.

### 2.3 Architecture, Inductive Bias, and Text Conditioning

Inductive bias refers to the assumptions a model makes about the structure of the data. In the context of PDEs, different architectures encode different inductive biases that influence their performance. Convolutional neural networks (CNNs), for instance, assume local spatial correlations and translation invariance, which align well with many physical systems. This makes them particularly effective for grid-based PDE problems.

On the other hand, transformer-based models rely on attention mechanisms that capture global dependencies. While this can be advantageous for certain tasks, it may not always align with the local nature of differential operators in PDEs. Additionally, transformers typically require larger datasets and longer training times to achieve comparable performance.

Multimodal learning combines different sources of information to improve prediction. In PDE modeling, text can provide structured context about the governing equation, boundary conditions, and qualitative system behavior. In this study, text is treated as metadata that complements numerical input rather than replaces it.

The motivation for this work arises from the intersection of these observations. Data-driven PDE solvers still struggle with OOD behavior, and textual descriptions provide rich contextual information that is currently underused in scientific machine learning. This work therefore tests whether text-based conditioning can improve OOD behavior in PDE surrogate models. At the same time, it treats multimodal learning as an architectural question rather than a guaranteed benefit: if text is useful, the model still needs a way to fuse semantic and spatial-temporal information effectively. Comparing convolutional and transformer-based models therefore helps clarify whether performance depends more on additional context or on the underlying inductive bias of the solver.

# 3 Related Work

## 3.1 Classical, Neural, and Operator-Based PDE Solvers

Classical PDE solvers such as finite difference, finite element, and spectral methods remain the standard tools because they are accurate and physically grounded. Their drawback is cost: they can be slow to run repeatedly and often need careful hand-designed discretizations for each new problem.

Neural surrogate models try to learn a shortcut. Instead of solving the equations from scratch every time, they learn from many examples and then predict the next state quickly. Early neural approaches such as physics-informed neural networks (PINNs) add the PDE itself to the loss so the network is punished when it violates known physics (Raissi et al., 2019; Wuwu et al., 2025). Operator-learning methods such as DeepONet and Fourier Neural Operator (FNO) go one step further: rather than memorizing one trajectory, they learn a mapping from one function to another (Lu et al., 2021; Li et al., 2021). In simple terms, they try to learn the rule of the game, not just one game result.

Another closely related idea is message passing on spatial grids or graphs. Message Passing Neural PDE Solvers (MPP) learn how nearby spatial locations exchange information over time (Brandstetter et al., 2022). This is a natural fit for PDEs because many physical laws are local: what happens at one point usually depends most strongly on its neighbors. These families of models are fast after training, but they can still struggle when the test problem looks meaningfully different from the training problems.

## 3.2 Transformer-Based PDE Solvers

Transformer-based PDE models ask whether a more flexible architecture can generalize across many physical systems by combining field information with more structured global context.

Unisolver is a representative example (Zhou et al., 2024). It conditions the model on structured PDE ingredients such as equation terms, coefficients, and boundary conditions. This is important because two systems can exhibit similar instantaneous states while still evolving differently under distinct governing equations or boundary constraints. By incorporating these structured descriptors explicitly, the model is better positioned to separate visual similarity from physical equivalence.

A second direction is large-scale pretraining across many physical systems. Multi-physics pretraining (MPP) aims to expose a model to diverse simulation families so that it learns reusable physical patterns before being adapted to a specific task (Brandstetter et al., 2022). This is attractive because PDE generalization usually fails when a model only knows one narrow regime. However, transformer-style models also bring costs: they often need more data, more compute, and stronger regularization than convolutional baselines, and they may be less naturally aligned with strongly local operators.

### 3.3 Text-Conditioned and Multimodal PDE Models

Recent work has started to add language directly to PDE modeling. Lorsung and Barati Farimani study whether pretrained language embeddings can help surrogate models when the text describes the system being solved (Lorsung and Barati Farimani, 2024). Text2PDE explores a more generative setup, where language is used to guide physics simulation through latent diffusion (Zhou et al., 2025). In both cases, the central idea is simple: a short text description can summarize things that may be hard to infer from raw fields alone, such as the type of equation, the forcing, or the boundary behavior.

This is promising, but it is not automatic. A text embedding is only useful if the numerical model knows how to use it. If the fusion mechanism is weak, the text becomes just another vector attached to the input and may be ignored. That is why multimodal PDE learning is still an open problem rather than a solved recipe.

### 3.4 Symbolic, Hybrid, and LLM-Guided Directions

Another branch of the literature uses language-model-style systems more directly for symbolic reasoning. CodePDE, for example, frames PDE solving as an inference and generation problem in which the model helps construct solver logic rather than only forecasting states (Li et al., 2025). Hybrid work in this area combines neural prediction, symbolic structure, and physics constraints to improve interpretability or automation.

These methods are exciting, but they answer a somewhat different question from the one studied here. They focus on generating solver procedures or symbolic forms, whereas this report focuses on rollout prediction for grid-based surrogate models.

### 3.5 Research Gap and Positioning

The literature now covers strong convolutional surrogates, operator learners, transformer-based universal solvers, and early text-conditioned systems. The missing piece is still the interaction between semantic conditioning and architectural bias. We do not yet know whether a simple text vector helps a local convolutional solver generalize better, or whether text only becomes useful when paired with richer fusion mechanisms such as cross-attention.

This report targets that gap. It keeps the conditioning mechanism intentionally lightweight, applies it to a UNet-style surrogate, and compares the result against both an unconditioned UNet and representative AViT transformer runs. That makes the contribution narrower than a foundation-model paper, but clearer: it isolates whether simple text conditioning helps OOD PDE rollout prediction in a practical grid-based setting.

## 4 Methodology

### 4.1 Problem Formulation

We consider the problem of learning a data-driven surrogate model for solving PDEs. Let the physical system be described by a PDE of the form:

$$\mathcal{F}(u, \nabla u, \nabla^2 u, x, t; \theta) = 0$$

where $u(x, t)$ represents the system state defined over spatial domain $x \in \Omega$ and time $t \in \mathbb{R}^+$, and $\theta$ denotes system-specific parameters such as coefficients and boundary conditions.

The goal is to learn a parameterized evolution map that approximates one-step dynamics:

$$\hat{u}_{t+1} = f_\phi(u_{t-T_i+1:t}, c)$$

where: - $u_{t-T_i+1:t}$ denotes the sequence of $T_i$ input timesteps, - $c$ represents the projected text-conditioning vector, - $f_\phi$ is a neural network parameterized by $\phi$.

### 4.2 Data Representation

Each PDE state is represented as a multi-channel tensor: $u_t \in \mathbb{R}^{C \times H \times W}$, where $C$ denotes the number of physical variables and $H, W$ denote the spatial resolution.

The input to the model is constructed by stacking past timesteps along the channel dimension:

$$X_t = [u_{t-T_i+1}, \dots, u_t] \in \mathbb{R}^{(T_i C)\times H\times W}.$$

The target output is:

$$Y_t = [u_{t+1}, \dots, u_{t+T_o}] \in \mathbb{R}^{(T_o C)\times H\times W}.$$

### 4.3 Text-Based Conditioning

To incorporate high-level physical context, we use natural language descriptions of PDE systems. Each description includes:

- Governing equations
- Boundary conditions
- Operator coefficients
- Qualitative system behavior

These descriptions are processed using the frozen SentenceTransformer model `sentence-transformers/all-MiniLM-L6-v2`. Let $s$ denote the text description and $g_{\text{MiniLM}}$ denote this frozen encoder. The resulting embeddings are cached in `pde_text_embeddings.pt` and reused during training:

$$e = g_{\text{MiniLM}}(s), \qquad e \in \mathbb{R}^{384}.$$

The embedding is then projected into a conditioning vector using a feedforward network:

$$c = W_2 \sigma(W_1 e + b_1) + b_2$$

where:

- $\sigma$ is the SiLU activation function,
- $W_1 \in \mathbb{R}^{128\times 384}$ and $W_2 \in \mathbb{R}^{16\times 128}$ are learnable parameters,
- $c \in \mathbb{R}^{16}$ is the projected conditioning vector.

### 4.4 Text Conditioning Descriptions

The conditioning descriptions used in this work follow a compact four-layer schema before embedding with `sentence-transformers/all-MiniLM-L6-v2`:

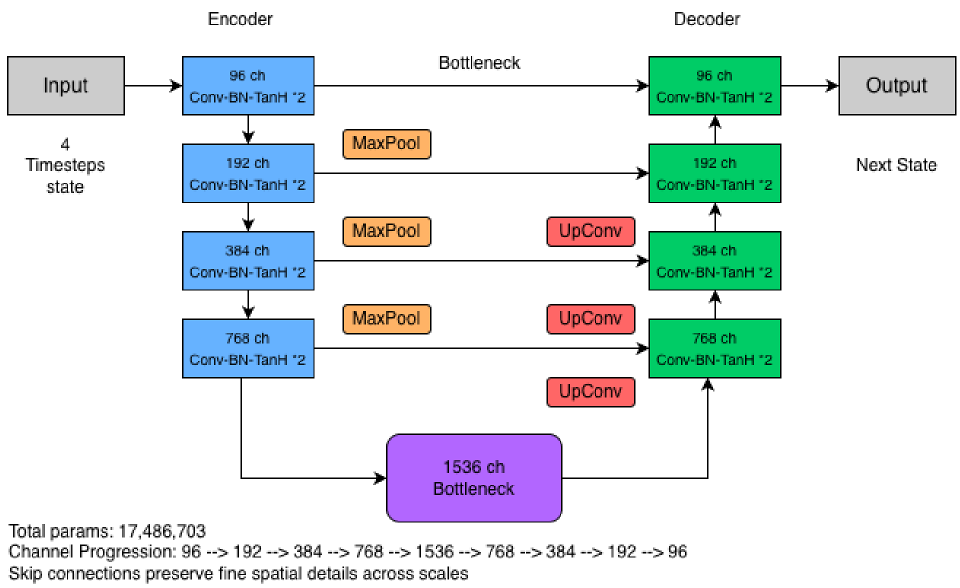
1. Basic description: outlines the equation type and core physical properties.
2. Boundary conditions: adds constraint information at the domain boundaries.
3. Operator coefficients: specifies the numerical parameters that appear in the PDE.
4. Qualitative behavior: captures intuitive dynamics that may not be written explicitly in the equation.

**Turbulent radiative layer.** Turbulent radiative layer in two spatial dimensions governed by compressible Navier–Stokes equations coupled with thermal energy transport. The system includes nonlinear advection, viscous diffusion, and buoyancy forcing driven by vertical temperature gradients. Lower boundary injects heat, upper boundary removes heat through radiative cooling, horizontal boundaries are periodic. Flow regime is convection driven and turbulence dominated. The solution develops multi-scale vortices, thermal plumes, and chaotic velocity structures. The predicted state should exhibit intermittent turbulent mixing and filamentary temperature fields.

**Active matter.** Two-dimensional active matter system governed by nonlinear hydrodynamic equations with self-propulsion forcing and diffusive regularization. The velocity field includes advective transport and active energy injection at small scales. Periodic boundary conditions in both spatial directions. The system is far from equilibrium and activity dominated. The solution develops spontaneous vortices, traveling bands, and long-range collective motion. The predicted state should exhibit coherent swirling structures and dynamically forming clusters.

### 4.5 UNet-Based Architecture

Figure 3: UNet architecture used in this work.




The backbone of our model is a UNet architecture, which consists of an encoder–decoder structure with skip connections as shown in Figure 3.

#### 4.5.1 Encoder

The encoder progressively extracts hierarchical features:

$$h_1 = B_1(X_t), \qquad h_l = B_l(\mathrm{Pool}(h_{l-1})), \quad l = 2, \dots, 4.$$

Each block consists of two repeated convolutional stages with batch normalization and Tanh activations.

The spatial resolution is reduced using max pooling, while the number of channels increases:

$$48 \to 96 \to 192 \to 384 \to 768$$

#### 4.5.2 Bottleneck

At the lowest resolution, the model captures global context:

$$h_\mathrm{b} = B_\mathrm{b}(\mathrm{Pool}(h_4))$$

#### 4.5.3 Decoder

The decoder reconstructs the output through upsampling:

$$\tilde{h}_l = D_l(\mathrm{Concat}_\mathrm{ch}(\mathrm{Up}(\tilde{h}_{l+1}), h_l)), \quad l = 4, \dots, 1.$$

where: - $\mathrm{Up}(\cdot)$ denotes transposed-convolution upsampling, - $h_l$ denotes the skip connection from the encoder.

### 4.6 Conditioning Integration

The conditioning vector $c$ is integrated into the model by augmenting the input representation:

$$X'_t = \text{Concat}_{\text{ch}}(X_t, c^{\uparrow}), \qquad c^{\uparrow} \in \mathbb{R}^{16\times H\times W}.$$

where $c^{\uparrow}$ is a spatially broadcasted version of the conditioning vector.

This allows the model to adapt its predictions based on textual context.

### 4.7 Training Objective

The model is trained using supervised learning with a mean squared error (MSE) loss:

$$L_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N} \|\hat{u}_i - u_i\|_2^2$$

To evaluate robustness, we also consider rollout loss over multiple timesteps:

$$L_{\text{rollout}} = \frac{1}{T_r}\sum_{\tau=1}^{T_r} L_{\text{MSE}}(\hat{u}_{t+\tau}, u_{t+\tau})$$

### 4.8 Training Procedure

At each iteration, the model samples an input-target pair, computes the text embedding and projected conditioning vector, predicts the next state, and updates parameters using the training loss.

### 4.9 Evaluation Protocol

We evaluate models using multi-step rollout:

- Predict one step ahead
- Feed prediction back as input
- Repeat for multiple timesteps

Metrics used:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Maximum error ($L_\infty$)

This setup tests both short-term accuracy and long-term stability.

### 4.10 Baselines

We compare:

- UNetClassic (unconditioned baseline)
- UNetClassicConditioned (text-conditioned baseline)
- Axial Vision Transformer (AViT) (transformer baseline)

### 4.11 Methodological Design Choices

Overall, the methodology combines spatial modeling via a UNet backbone, temporal modeling via stacked input frames, and semantic conditioning via text embeddings. The central design question is whether this relatively simple fusion strategy is strong enough to improve OOD PDE surrogate modeling without disrupting the inductive bias that makes convolutional models effective on grid-structured physical fields.

## 5 Experiments and Results

### 5.1 Model Implementation

The models were implemented in the project repository using a GPU-accelerated deep learning pipeline. The UNet baseline and the conditioned UNet follow the methodology described above; in practice, the conditioned variant adds the frozen `sentence-transformers/all-MiniLM-L6-v2` embedding pathway and projects text features from $\mathbb{R}^{384} \to \mathbb{R}^{128} \to \mathbb{R}^{16}$ before broadcasting them across the spatial grid.

The repository currently exposes a single AViT configuration without a separate text-conditioning path. For that reason, the AViT results reported below should be interpreted as representative transformer-baseline runs rather than as a matched conditioned-versus-unconditioned AViT comparison.

### 5.2 Dataset and Preprocessing

The experiments were conducted on multiple PDE datasets:

- Training datasets:
  - Turbulent Radiative Layer (2D)
  - Active Matter
- Validation datasets:
  - Turbulent Radiative Layer (2D)
  - Active Matter
- Test dataset (OOD):
  - Helmholtz Staircase

Each dataset consists of spatiotemporal simulations represented as multi-channel tensors. The spatial resolution is fixed at $128 \times 384$, and temporal sequences are constructed using:

- Input timesteps: 4
- Output timesteps: 1

Input sequences are formed by stacking consecutive frames along the channel dimension. In the multi-dataset setting, all fields are resized to a common $128 \times 384$ grid and remapped into a shared channel schema so that models can process turbulent radiative layer, active matter, and Helmholtz staircase data with the same input and output dimensionality. All data is normalized to ensure stable training.

### 5.3 Training Configuration

The models were trained using the following setup:

- Optimizer: AdamW
- Learning rate: $10^{-2}$
- Weight decay: $10^{-4}$
- Learning rate scheduler: Linear warmup followed by cosine annealing
- Loss function: Mean Squared Error (MSE)
- Evaluation: Multi-step rollout
- Number of epochs: 20
- Batch size: 8

Training was performed for a fixed number of epochs, and performance was evaluated on both in-distribution and out-of-distribution datasets. The in-distribution (ID) validation set uses the same PDE families as training, while the Helmholtz staircase test set is reserved for OOD evaluation. The headline values in Table 1 and Table 2 are taken from a consistent set of saved experiment summaries: `UNetClassic-0.01/6`, `UNetClassicConditioned-0.01/5`, and two representative `AViT-0.01` runs (`/9` and `/8`) from the multi-dataset experiment folder.

Table 1: In-distribution and OOD performance.

| Model | ID Val. Loss | OOD Test Loss | Gen. Gap |
|---|---|---|---|
| UNetClassic (baseline) | 0.00948 | 0.13623 | 0.12675 |
| UNetClassicConditioned | 0.00982 | 0.13802 | 0.12820 |
| AViT (representative run A) | 0.01134 | 0.41496 | 0.40362 |
| AViT (representative run B) | 0.01146 | 0.48020 | 0.46873 |

Table 2: OOD rollout metrics on Helmholtz staircase.

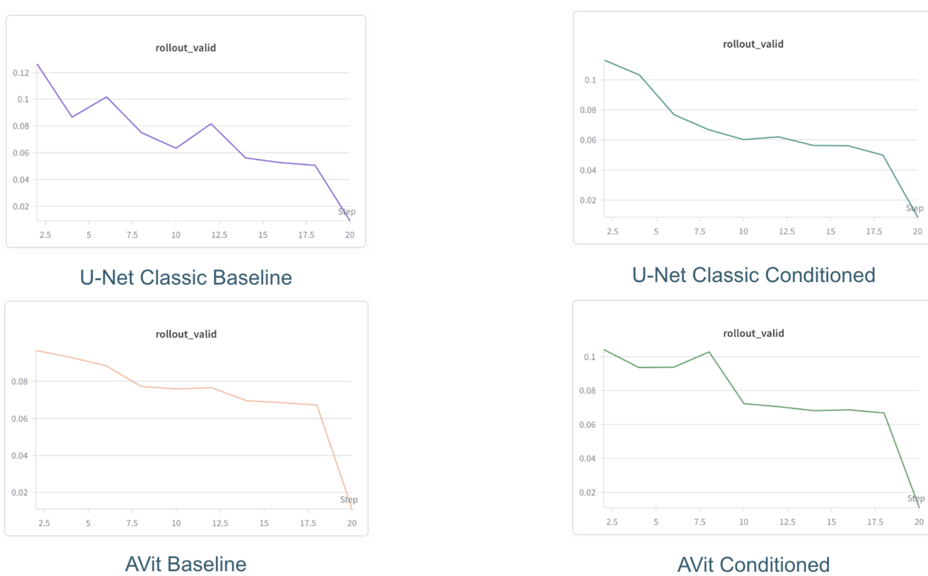
| Model | OOD Rollout Test Loss | MSE ($D_{xx}$) | RMSE | $L_\infty$ Error |
|---|---|---|---|---|
| UNetClassic (baseline) | 0.13623 | 0.00096 | 0.03091 | 0.18391 |
| UNetClassicConditioned | 0.13802 | 0.00637 | 0.07981 | 0.23783 |
| AViT (representative run A) | 0.41496 | 0.18382 | 0.40078 | 1.48433 |
| AViT (representative run B) | 0.48020 | 0.23509 | 0.45017 | 1.51812 |

### 5.4 Overall Performance

Because OOD generalization is the main research question in this work, Table 1 explicitly separates in-distribution and out-of-distribution results. The table compares validation loss on the in-distribution validation set with rollout test loss on the Helmholtz staircase OOD test set, and the final column reports the corresponding generalization gap.

All models achieve relatively low in-distribution validation loss, but performance degrades substantially on the OOD Helmholtz staircase test set. The baseline UNet retains the smallest degradation, while the conditioned UNet and the two representative AViT runs exhibit markedly larger generalization gaps.

Figure 4: Rollout loss on the Helmholtz staircase OOD evaluation.




As shown in Figure 4, the rollout loss curve highlights that the baseline UNetClassic model accumulates error more slowly than both the conditioned UNet and the reported AViT runs. This visual trend supports the numerical results in Table 2, which report OOD-only rollout metrics on the Helmholtz staircase test set.

The results for $T_i = 4$ are presented in Table 2. The UNetClassic model (without embedding) achieves the best OOD performance, with the lowest rollout loss and the most favorable error metrics across all evaluations. The conditioned variant (UNetClassicConditioned) performs slightly worse, indicating that text embedding does not improve performance in this setting. The two representative AViT runs perform substantially worse than the UNet-based models, with OOD rollout losses between 0.41496 and 0.48020. This again suggests that architectural suitability matters more than simply switching to a transformer baseline.

### 5.5 Field-wise Error Analysis

A detailed evaluation on individual fields (e.g., $D_{xx}$) reveals:

- UNetClassic achieves the lowest error across all metrics
- UNetClassicConditioned shows significant degradation:
  - approximately 6.7× higher MSE
  - approximately 1.3× higher $L_\infty$ error
- AViT models exhibit the highest errors across all metrics

These results indicate that the baseline model is more effective at capturing the underlying dynamics.

### 5.6 Parameter Norm Analysis

The parameter norms of the representative checkpoints are:

- UNetClassic: ~2597
- UNetClassicConditioned: ~2924
- AViT: ~3850+

Larger parameter norms in the AViT checkpoints do not translate into better OOD performance. This suggests that raw parameter magnitude is not the limiting factor and again highlights the importance of architectural suitability.

### 5.7 Model Complexity

To ensure a fair comparison between architectures, we report the total number of trainable parameters for each model:

- UNetClassic: 17,512,623 parameters
- UNetClassicConditioned: 17,570,879 parameters
- AViT (representative runs A and B): 14,841,856 parameters each

The conditioned UNet model introduces a marginal increase in parameters due to the additional projection layers used for incorporating text embeddings. The AViT architecture keeps the same parameter count across the reported runs because the repository currently exposes a single AViT configuration.

Despite having fewer parameters, AViT models underperform compared to UNet-based architectures, indicating that performance differences are primarily driven by architectural inductive bias rather than model size.

### 5.8 Effect of Text Conditioning

One of the key findings of this work is that text conditioning does not improve performance in the current setup. While the hypothesis suggested that textual information could enhance generalization, the experimental results indicate otherwise.

Several factors contribute to this outcome:

- Redundant information: The multi-dataset training setup already encodes dataset identity through channel structure, reducing the need for additional conditioning.
- Weak alignment between modalities: The text embeddings may not align effectively with the numerical features learned by the UNet.
- Overfitting to conditioning signal: The additional parameters introduced by the conditioning module may lead to overfitting without improving generalization.

These observations suggest that simply adding text embeddings is insufficient, and more sophisticated integration strategies are required.

### 5.9 Architectural Considerations

The comparison between UNet and AViT highlights the importance of inductive bias in model design.

- UNet advantages:
  - Strong inductive bias for grid-based data
  - Efficient local feature extraction
  - Multi-scale representation through pooling
- AViT limitations:
  - Attention mechanisms are not well-suited for local PDE operators
  - Requires larger datasets and longer training
  - Higher computational complexity

As a result, the UNet architecture is better aligned with the structure of PDE data, leading to superior performance.

### 5.10 Generalization and OOD Performance

The evaluation on the Helmholtz staircase dataset demonstrates that generalization remains a challenging problem. While the baseline UNet performs well, the conditioned model does not show improved OOD performance.

This gap is also visible when comparing in-distribution and out-of-distribution losses directly. In one representative run, the training loss decreases from approximately 1.05 to 0.0069 by epoch 20, while the validation loss remains low at about 0.0103. In contrast, the final test loss rises to 0.1398. This indicates that the model fits the training and validation distributions well, but degrades substantially on the OOD test setting. Since Helmholtz staircase is used as the OOD benchmark, this separation between ID and OOD performance should be made explicit in the interpretation of the results.

This suggests that:

- The conditioning signal is not effectively guiding the model
- The model still relies primarily on learned numerical patterns

Improving OOD generalization requires:

- Better fusion mechanisms
- More expressive representations of physical knowledge

### 5.11 Implications for Multimodal PDE Modeling

The results of this work provide important insights into multimodal learning for scientific applications:

- Multimodal inputs can be beneficial, but integration strategy is critical
- Architectural alignment is more important than simply increasing model complexity
- Text-based conditioning requires careful design to be effective

These findings highlight the need for future research in:

- Cross-modal attention mechanisms
- Physics-aware embedding strategies
- Hybrid architectures combining CNNs and transformers

### 5.12 Discussion

The implementation and evaluation together demonstrate that:

- UNet-based models are highly effective for PDE surrogate modeling
- Text conditioning, in its current form, does not improve performance
- Transformer-based models underperform due to architectural mismatch

The results emphasize that model design and modality integration are key factors in developing robust and generalizable PDE solvers.

## 6 Limitations

This study has several limitations that are important for interpreting the results. First, the evaluation is restricted to a small set of PDE families, with Helmholtz staircase serving as the primary OOD benchmark. While this is sufficient to probe cross-domain generalization, it does not fully establish how the proposed conditioning strategy would behave across a broader range of equations, resolutions, and physical regimes.

Second, the text-conditioning mechanism is deliberately simple: a frozen language-model embedding is projected and concatenated to the numerical input. This design is useful for isolating the question of whether lightweight semantic conditioning helps, but it may be too weak to fully align language information with spatial-temporal solver dynamics. A stronger fusion mechanism, such as cross-attention or intermediate feature conditioning, might yield different conclusions.

Third, the transformer comparison should be interpreted carefully. In the current repository, the reported Axial Vision Transformer runs are representative AViT baselines rather than a matched pair of conditioned and unconditioned AViT models. As a result, the comparison is informative about architectural differences, but it is less definitive about whether text conditioning would help within a more fully multimodal transformer design.

## 7 Conclusion

This work evaluated whether text-based descriptions can improve data-driven PDE surrogate modeling. A conditioned UNet was developed by projecting structured natural language descriptions into embeddings and incorporating them alongside numerical inputs. The model was compared with an unconditioned UNet and transformer-based AViT baselines on turbulent radiative layer, active matter, and Helmholtz staircase datasets using rollout-based error metrics.

The results show that the baseline UNet remains the strongest model in this setting. Text conditioning does not consistently improve performance and can degrade accuracy, while the transformer baseline underperforms despite having a comparable overall model scale. These outcomes suggest that architectural inductive bias matters more than model capacity alone for the grid-based PDE systems studied here.

The main lesson is that multimodal PDE modeling requires more than appending a text embedding to numerical states. The conditioning mechanism must align semantic information with spatial and temporal dynamics in a way the solver can use. Future work should therefore prioritize stronger fusion strategies, such as cross-attention or physics-aware embeddings, and hybrid architectures that preserve the locality advantages of convolutional models while allowing richer conditioning.

Overall, the study provides a systematic negative result: simple text conditioning is not sufficient for improved OOD generalization, but it clarifies where future multimodal PDE solvers need better design.

## 8 Reproducibility Statement

Code and experiment setup are available at Divyamdj/pde_solver. The exact training configuration, dataset split, and evaluation setup reported in this paper are described in the `Experiments and Results` section. Text embeddings are generated once using the frozen `sentence-transformers/all-MiniLM-L6-v2` encoder and cached in `pde_text_embeddings.pt`. The headline results in both tables are computed from the saved experiment summaries in `...UNetClassic-0.01/6`, `...UNetClassicConditioned-`

0.01/5, ...AViT-0.01/9, and ...AViT-0.01/8. No additional private data or human annotations are used.

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